

THE EQUAL-NORM CASE: NORM-SEPARATION IS NOT A HYPOTHESIS BUT A CHARACTERIZATION, AND THE GAUGE-EQUIVARIANT REPAIR

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ABSTRACT. Paper XIII of the series proved $\mathcal{A}_{K,S}^\sigma = C(Y_S)$ (its Theorem 1.4) under the hypothesis that $N : J_S \rightarrow \mathbb{N}$ be injective, and asked whether the hypothesis could be removed. It cannot, and the reason is that it is not a hypothesis at all.

The diagonal is always the fixed-point algebra of the *compact* gauge action: $C(Y_S) = \mathcal{A}_{K,S}^{\mathbb{T}^S}$, unconditionally. The modular flow is the restriction of the gauge action along $\iota(t)_p = Np^{it}$, so $\mathcal{A}^\sigma = \mathcal{A}^H$ with $H = \overline{\iota(\mathbb{R})} \leq \mathbb{T}^S$, and by Kronecker $H = \mathbb{T}^S$ exactly when the $\log Np$ are $\mathbb{Q}$ -linearly independent. Since $\mathbb{Z}$ -independence of the $\log Np$ is literally the injectivity of N on J_S , we obtain

$$\mathcal{A}_{K,S}^\sigma = C(Y_S) \iff \{\log Np\}_{p \in S} \text{ } \mathbb{Q}\text{-independent} \iff S \text{ norm-separated.}$$

The three conditions are equivalent; there is nothing to remove.

The correct structure in the general case is not the proposed direct sum $\bigoplus_n \mathcal{A}_n^\sigma$, which is not a direct sum — the pieces are neither orthogonal nor ideals, since $e_a e_b = e_{a \vee b} \neq 0$ and the product of a degree- n and a degree- m element is again in $\mathcal{A}^\sigma$. It is a grading. Writing $L = \{c \in \mathbb{Z}^{(S)} : \sum_p c_p \log Np = 0\}$ for the lattice of multiplicative relations among the norms, $\mathcal{A}^\sigma$ is L -graded with degree-zero part $C(Y_S)$ and $H = L^\perp$.

Neither of the two proposed intrinsic characterizations of $C(Y_S)$ inside $\mathcal{A}^\sigma$ works as stated. Property A recovers $C(Y_S)$ as the fixed points of the gauge action, but the gauge action of $\mathbb{T}^S$ is precisely the datum *not* determined by $(\mathcal{A}, \sigma)$ when $L \neq 0$; the argument is circular. Property B asserts that $C(Y_S)$ is the unique Cartan subalgebra, which is what was to be proved.

What does work is a change of hypothesis rather than a removal. If Φ is equivariant for the full gauge actions — up to an isomorphism of the acting tori, so that no bijection $S_1 \rightarrow S_2$ is presupposed — then $\Phi(C(Y_{S_1})) = C(Y_{S_2})$ unconditionally, and the reconstruction of Paper XIII (its Theorem 2.5) holds for every number field and every prime set: a norm-preserving bijection $S_1 \rightarrow S_2$, which turns out to induce the torus isomorphism after all; the matched valuation fibres, with G_S as a compact space; the KMS_β simplices, transition loci and β_c ; and, when the symmetry actions are intertwined, the Frobenius classes modulo inertia and the chain $\{\Xi_\beta\}$. For $S_i = P_{K_i}$ the norm multisets agree, so $\zeta_{K_1} = \zeta_{K_2}$: the fields are arithmetically equivalent. The narrow class number is not among the conclusions; it was not recovered in Paper XIII either.

Finally we record a limitation of Paper XIII that we had not noticed there: every $K \neq \mathbb{Q}$ has infinitely many split primes, so P_K is *never* norm-separated. The theorem of Paper XIII therefore covers the classical full prime set only for $K = \mathbb{Q}$.

1. NORM-SEPARATION IS A CHARACTERIZATION

We use the notation of the foundational paper [3]: $\mathcal{A}_{K,S} = C(Y_S) \rtimes J_S$ is spanned by the elements $\mu_a f \mu_b^*$, and it is the C^* -algebra of the étale groupoid $\mathcal{G}_S|_{Y_S}$ whose arrows are the pairs (ab^{-1}, y) with $y, ab^{-1}y \in Y_S$, with diagonal $C(Y_S)$ and with the scaling cocycle $c(ab^{-1}, y) = \log Na - \log Nb$ implementing the flow. Papers II, III and XIII of the series, referred to by numeral, are in preparation; every fact used from them is re-proved in place. The finer, integer-valued cocycle

$$\tilde{c} : \mathcal{G}_S|_{Y_S} \longrightarrow I_S = \mathbb{Z}^{(S)}, \quad \tilde{c}(ab^{-1}, y) = ab^{-1},$$

Sources, code and data for the entire series: <https://github.com/Ruqing1963/localized-bost-connes>; this paper is in [papers/XIV-equalnorm/](#).

is continuous and locally constant, and composing it with characters defines the *gauge action* of the compact group $\mathbb{T}^S = \widehat{I}_S$:

$$\gamma_z(f)(\gamma) = z^{\tilde{c}(\gamma)} f(\gamma), \quad \text{so} \quad \gamma_z(\mu_{\mathfrak{p}}) = z_{\mathfrak{p}} \mu_{\mathfrak{p}}, \quad \gamma_z|_{C(Y_S)} = \text{id}, \quad z \in \mathbb{T}^S,$$

strongly continuous because $\tilde{c}$ is locally constant. Since $c = \langle \tilde{c}, \log N \rangle$, the modular flow is a restriction of the gauge action: $\sigma_t = \gamma_{\iota(t)}$ with $\iota(t)_{\mathfrak{p}} = N\mathfrak{p}^{it}$ (the identity through which Paper III of the series sees that $\{\sigma_t\}$ has compact closure).

Lemma 1.1 (The diagonal is the gauge fixed-point algebra). $\mathcal{A}_{K,S}^{\mathbb{T}^S} = C(Y_S)$, for every K and every S , with no hypothesis.

Proof. $\gamma_z(\mu_{\mathfrak{a}} f \mu_{\mathfrak{b}}^*) = z^{\mathfrak{a}-\mathfrak{b}} \mu_{\mathfrak{a}} f \mu_{\mathfrak{b}}^*$ in multi-index notation, so a spectral subspace argument gives $\mathcal{A}^{\mathbb{T}^S} = \overline{\text{span}}\{\mu_{\mathfrak{a}} f \mu_{\mathfrak{a}}^*\}$, and $\mu_{\mathfrak{a}} f \mu_{\mathfrak{a}}^* = \alpha_{\mathfrak{a}}(f) \in C(Y_S)$. $\square$

Definition 1.2. The *relation lattice* of S is

$$L = L(S) := \left\{ c \in \mathbb{Z}^{(S)} : \sum_{\mathfrak{p} \in S} c_{\mathfrak{p}} \log N\mathfrak{p} = 0 \right\} = \left\{ c : \prod_{\mathfrak{p}} N\mathfrak{p}^{c_{\mathfrak{p}}} = 1 \right\},$$

a subgroup of $\mathbb{Z}^{(S)}$.

Theorem 1.3 (Equivalence). *The following are equivalent.*

- (1) $N : J_S \rightarrow \mathbb{N}$ is injective, i.e. S is norm-separated;
- (2) $L(S) = 0$, i.e. $\{\log N\mathfrak{p}\}_{\mathfrak{p} \in S}$ is $\mathbb{Q}$ -linearly independent;
- (3) $\iota(\mathbb{R})$ is dense in $\mathbb{T}^S$;
- (4) $\mathcal{A}_{K,S}^{\sigma} = C(Y_S)$.

Proof. (1) $\Leftrightarrow$ (2): $N\mathfrak{a} = N\mathfrak{b}$ with $\mathfrak{a}, \mathfrak{b} \in J_S$ means $\sum (a_{\mathfrak{p}} - b_{\mathfrak{p}}) \log N\mathfrak{p} = 0$ with $a - b \in \mathbb{Z}^{(S)}$; injectivity says the only such relation is $a = b$, i.e. $L = 0$. A $\mathbb{Q}$ -linear relation clears denominators to a $\mathbb{Z}$ -linear one, so $\mathbb{Z}$ -independence and $\mathbb{Q}$ -independence agree.

(2) $\Leftrightarrow$ (3): $\overline{\iota(\mathbb{R})}$ is a closed subgroup of $\mathbb{T}^S$, and its annihilator in $\widehat{\mathbb{T}^S} = \mathbb{Z}^{(S)}$ is $\{c : \prod_{\mathfrak{p}} N\mathfrak{p}^{ic_{\mathfrak{p}}} = 1 \ \forall t\} = L$. By Pontryagin duality $\overline{\iota(\mathbb{R})} = \mathbb{T}^S$ iff $L = 0$; this is Kronecker's theorem.

(3) $\Leftrightarrow$ (4): $\mathcal{A}^{\sigma} = \overline{\mathcal{A}^{\iota(\mathbb{R})}}$, since the fixed-point algebra of a one-parameter group equals that of the closure of its image; combine with Lemma 1.1. Conversely if $L \neq 0$, pick $0 \neq c \in L$ and write $c = \mathfrak{a} - \mathfrak{b}$ with $\mathfrak{a}, \mathfrak{b} \in J_S$ coprime; then $N\mathfrak{a} = N\mathfrak{b}$ and $\mu_{\mathfrak{a}} \mu_{\mathfrak{b}}^* \in \mathcal{A}^{\sigma}$ is not in $C(Y_S)$, the two ideals being distinct. $\square$

Corollary 1.4 (Nothing to remove). *The hypothesis of Theorem 1.4 of Paper XIII is not a sufficient condition that might be weakened: it is equivalent to the conclusion. In the unseparated case $\mathcal{A}^{\sigma}$ is strictly larger than $C(Y_S)$ and non-commutative, and no argument can identify them.*

Proposition 1.5 (P_K is never separated for $K \neq \mathbb{Q}$). *Let $K \neq \mathbb{Q}$. By the Chebotarev density theorem [2, Ch. VII, §13], applied to the Galois closure of K , there are infinitely many rational primes that split completely in K ; the $[K : \mathbb{Q}] \geq 2$ primes of K above such a p have residue degree 1 and hence equal norm p . So P_K is not norm-separated, and Theorem 1.4 of Paper XIII applies to the full prime set only when $K = \mathbb{Q}$.*

Remark 1.6 (Numerical illustration). The first split prime is 5 for $\mathbb{Q}(i)$, 7 for $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(\sqrt{2})$, and 11 for $\mathbb{Q}(\sqrt{5})$; in each case two primes of norm p . This is a limitation of Paper XIII that we had not noticed there: the classical case, $S = P_K$ over a general number field, is precisely the case the theorem does not cover.

2. THE STRUCTURE OF $\mathcal{A}^\sigma$ IN GENERAL

Proposition 2.1 (The proposed stratification is not a direct sum). *Set $\mathcal{A}_n^\sigma = \overline{\text{span}}\{\mu_{\mathbf{a}}f\mu_{\mathbf{b}}^* : N\mathbf{a} = N\mathbf{b} = n\}$. Then $\mathcal{A}^\sigma \neq \bigoplus_n \mathcal{A}_n^\sigma$: the summands are not orthogonal, since $e_{\mathbf{a}}e_{\mathbf{b}} = e_{\text{lcm}(\mathbf{a},\mathbf{b})} \neq 0$ for $\mathbf{a}, \mathbf{b}$ of different norms, and they are not ideals, the product of an element of $\mathcal{A}_n^\sigma$ with one of $\mathcal{A}_m^\sigma$ lying in $\mathcal{A}^\sigma$ but in neither.*

Proof. $e_{\mathbf{a}} = \mu_{\mathbf{a}}\mu_{\mathbf{a}}^* \in \mathcal{A}_{N\mathbf{a}}^\sigma$ and $e_{\mathbf{a}}e_{\mathbf{b}} = e_{\text{lcm}} \neq 0$; for the second, compute $(\mu_{\mathbf{a}}f\mu_{\mathbf{b}}^*)(\mu_{\mathbf{c}}g\mu_{\mathbf{d}}^*)$ using $\mu_{\mathbf{b}}^*\mu_{\mathbf{c}} = \mu_{\mathbf{c}'}\mu_{\mathbf{b}'}^*$ with $\mathbf{c}' = \mathbf{c}/\text{gcd}$, $\mathbf{b}' = \mathbf{b}/\text{gcd}$, obtaining a term $\mu_{\mathbf{ac}'}(\cdots)\mu_{\mathbf{db}'}^*$ whose two norms are equal but need not be n or m . $\square$

Theorem 2.2 (The correct structure). $\mathcal{A}^\sigma = \mathcal{A}^H$ with $H = L^\perp \leq \mathbb{T}^S$ the annihilator of the relation lattice, and $\mathcal{A}^\sigma$ carries a grading by L ,

$$\mathcal{A}^\sigma = \bigoplus_{c \in L} \mathcal{A}_{(c)}^\sigma, \quad \mathcal{A}_{(c)}^\sigma = \overline{\text{span}}\{\mu_{\mathbf{a}}f\mu_{\mathbf{b}}^* : \mathbf{a} - \mathbf{b} = c\}, \quad \mathcal{A}_{(0)}^\sigma = C(Y_S),$$

the grading being implemented by the residual gauge action of $\mathbb{T}^S/H \cong \widehat{L}$. In particular $C(Y_S) = (\mathcal{A}^\sigma)^{\widehat{L}}$.

Proof. $H = \overline{i(\mathbb{R})}$ has annihilator L by the proof of Theorem 1.3, so $\mathbb{T}^S/H \cong \widehat{L}$ and the gauge action descends. The grading is the spectral decomposition for that residual action, the degree of $\mu_{\mathbf{a}}f\mu_{\mathbf{b}}^*$ being $\mathbf{a} - \mathbf{b} \in \mathbb{Z}^{(S)}$, which lies in L precisely on $\mathcal{A}^\sigma$. Degree zero is Lemma 1.1. $\square$

Example 2.3. Let $K = \mathbb{Q}(i)$ and $S = \{\mathbf{p}, \bar{\mathbf{p}}\}$ the two primes above 5. Then $L = \mathbb{Z}(1, -1)$, $\widehat{L} \cong \mathbb{T}$, and $\mathcal{A}^\sigma$ is $\mathbb{Z}$ -graded, the piece of degree $k(1, -1)$ being spanned by the elements

$$\mu_{\mathbf{p}}^{m+k}\mu_{\bar{\mathbf{p}}}^n f(\mu_{\mathbf{p}}^m\mu_{\bar{\mathbf{p}}}^{n+k})^*, \quad \text{for instance} \quad \mu_{\mathbf{p}}^k f\mu_{\bar{\mathbf{p}}}^{k*},$$

both sides having the same norm 5^{m+n+k} . The degree-zero part is $C(Y_S)$ and the whole is a crossed-product-like algebra over it; it is non-commutative, and no spectral argument using σ alone can see inside it.

3. WHY THE TWO PROPOSED CHARACTERIZATIONS ARE CIRCULAR

Remark 3.1 (Property A). The proposal is to characterize $C(Y_S)$ as the unique MASA of $\mathcal{A}^\sigma$ carrying a faithful conditional expectation invariant under the gauge symmetries $\mathbb{T}^S$. By Lemma 1.1 this is correct as a *description* — $C(Y_S)$ is the range of the averaging expectation E_γ — but it is not an intrinsic characterization inside $(\mathcal{A}, \sigma)$, because the $\mathbb{T}^S$ -action is exactly the datum that σ fails to determine when $L \neq 0$. By Theorem 1.3(3), σ determines only $H = L^\perp \subsetneq \mathbb{T}^S$. Invoking γ to recover $C(Y_S)$ therefore assumes what is missing.

Remark 3.2 (Property B). The proposal is to characterize $C(Y_S)$ as the unique abelian MASA whose normalizer generates $\mathcal{A}_{K,S}$, i.e. as the unique Cartan subalgebra. That $C(Y_S)$ is a Cartan subalgebra is Theorem 2.5 of Paper II of the series (topological principality; the argument is reproduced inside the proof of Theorem 4.2 below). That it is the *unique* one is precisely the statement needed to apply Renault's theorem [1] to a merely σ -equivariant isomorphism, which is what Theorem 1.4 of Paper XIII was designed to supply and what is at issue here. Assuming it is circular. We add that uniqueness of Cartan subalgebras is false for general C^* -algebras and is a hard theorem where it holds, so it cannot be assumed lightly.

Remark 3.3 (What we do not know). We have not found an intrinsic characterization of $C(Y_S)$ inside $(\mathcal{A}_{K,S}, \sigma)$ in the unseparated case, and we have not found an obstruction either. The natural attempt — to exhibit a σ -commuting automorphism moving $C(Y_S)$, which would settle it negatively — fails for a reason worth recording. In a Cuntz algebra the quasi-free rotation

$\mu_i \mapsto \sum_j u_{ji} \mu_j$ is an automorphism because $\mu_i^* \mu_j = \delta_{ij}$; here the isometries commute and satisfy $\mu_p^* \mu_q = \mu_q \mu_p^*$ for coprime p, q , so, for $|\alpha|^2 + |\beta|^2 = 1$ with $\alpha\beta \neq 0$, $(\alpha\mu_p + \beta\mu_q)^*(\alpha\mu_p + \beta\mu_q) = 1 + \bar{\alpha}\beta\mu_q\mu_p^* + \alpha\bar{\beta}\mu_p\mu_q^* \neq 1$ and the rotated element is not an isometry. The Nica–Toeplitz relations are rigid enough to block the obvious counterexample, which is a reason to think the question may have a positive answer; we have not found it.

4. THE GAUGE-EQUIVARIANT REPAIR

Definition 4.1. Call $\Phi : \mathcal{A}_{K_1, S_1} \rightarrow \mathcal{A}_{K_2, S_2}$ *gauge-equivariant* if there is a topological group isomorphism $\theta : \mathbb{T}^{S_1} \rightarrow \mathbb{T}^{S_2}$ with $\Phi \circ \gamma_z^1 = \gamma_{\theta(z)}^2 \circ \Phi$ for all z . No bijection $S_1 \rightarrow S_2$ is presupposed, and none is encoded in θ by itself: the automorphisms of $\mathbb{T}^S$ are dual to the automorphisms of the free abelian group $\mathbb{Z}^{(S)}$, most of which permute no coordinates. Theorem 4.2 produces a bijection, and shows that θ was induced by one after all.

Theorem 4.2 (Unconditional reconstruction). *Let K_1, K_2 be arbitrary number fields and S_1, S_2 arbitrary infinite prime sets, and let $\Phi : \mathcal{A}_{K_1, S_1} \rightarrow \mathcal{A}_{K_2, S_2}$ be a gauge-equivariant and σ -equivariant isomorphism. Then $\Phi(C(Y_{S_1})) = C(Y_{S_2})$, and:*

- (1) Φ is induced by an isomorphism θ_* of the underlying étale groupoids which intertwines the scaling cocycles, $c_2 \circ \theta_* = c_1$, and the integer-valued gauge cocycles, $\hat{\theta} \circ \tilde{c}_2 \circ \theta_* = \tilde{c}_1$, where $\hat{\theta} : \mathbb{Z}^{(S_2)} \rightarrow \mathbb{Z}^{(S_1)}$ is the isomorphism dual to θ ;
- (2) θ_* induces a norm-preserving monoid isomorphism $\pi : J_{S_1} \rightarrow J_{S_2}$, hence a norm-preserving bijection $S_1 \rightarrow S_2$; moreover $\hat{\theta}(\delta_{\pi(p)}) = \delta_p$, so θ is, a posteriori, the isomorphism induced by the bijection π of the coordinate circles;
- (3) the valuation fibres of Y_{S_1} are carried homeomorphically onto those of Y_{S_2} , compatibly with π ; in particular the fibre over 0, the compact space G_{S_1} , is carried homeomorphically onto G_{S_2} ;
- (4) the KMS_β simplices are affinely isomorphic for every β , so the compact spaces $G_{S_i}/\Xi_\beta^\perp$ are homeomorphic, the orders $|\Xi_\beta|$ agree when finite, the transition loci agree, and $\beta_c(S_1) = \beta_c(S_2)$;
- (5) if moreover Φ intertwines the symmetry actions along an isomorphism $\iota : G_{S_1} \rightarrow G_{S_2}$, then the Frobenius classes are matched modulo inertia, $\iota(\sigma_p) \in \sigma_{\pi(p)} r(\mathcal{O}_{\pi(p)}^\times)$, and $\hat{\iota}^{-1}$ carries $\Xi_\beta(K_2, S_2)$ onto $\Xi_\beta(K_1, S_1)$ for every $\beta > 0$.

These are the conclusions of Theorem 2.5 of Paper XIII, with no norm-separation required anywhere; what is conditional in (5) is conditional there too.

Proof. The Cartan pair. By Lemma 1.1, $C(Y_{S_i}) = \mathcal{A}_i^{\mathbb{T}^{S_i}}$, and gauge-equivariance, with θ surjective, gives $\Phi(\mathcal{A}_1^{\mathbb{T}^{S_1}}) = \mathcal{A}_2^{\mathbb{T}^{S_2}}$, i.e. $\Phi(C(Y_{S_1})) = C(Y_{S_2})$. Each pair $(C(Y_S) \subset \mathcal{A}_{K,S})$ is Cartan with trivial Weyl twist: the groupoid $\mathcal{G}_S|_{Y_S}$ is étale, Hausdorff, second countable and amenable, and it is topologically principal because the isotropy is trivial at every point with all valuations finite ($\mathbf{a}\mathbf{b}^{-1}y = y$ forces $\mathbf{a} = \mathbf{b}$ at such a point) and those points are dense — this reproduces Theorem 2.5(1) of Paper II of the series. Renault’s theorem [1] therefore provides an isomorphism $\theta_* : \mathcal{G}_{S_1}|_{Y_{S_1}} \rightarrow \mathcal{G}_{S_2}|_{Y_{S_2}}$ of étale groupoids inducing Φ .

Both cocycles are intertwined. For f supported on an open bisection B , $\Phi(f)$ is supported on $\theta_*(B)$ (up to the $\mathbb{T}$ -valued normalization of the trivialized twist, which cancels below), and γ_w^2 multiplies it there by $w^{\tilde{c}_2}$. Comparing $\Phi(\gamma_z^1 f) = \gamma_{\theta(z)}^2 \Phi(f)$ pointwise gives $z^{\tilde{c}_1(\gamma)} = \theta(z)^{\tilde{c}_2(\theta_* \gamma)}$ for all $z \in \mathbb{T}^{S_1}$, i.e. $\tilde{c}_1 = \hat{\theta} \circ \tilde{c}_2 \circ \theta_*$; comparing $\Phi(\sigma_t^1 f) = \sigma_t^2 \Phi(f)$ gives $c_2 \circ \theta_* = c_1$.

The semigroup, without norm-separation. A full bisection of $\mathcal{G}_S|_{Y_S}$ on which $\tilde{c}$ is constant contains, for each $y \in Y_S$, exactly one arrow (g, y) with $g = \mathbf{a}\mathbf{b}^{-1}$ independent of y , $\mathbf{a}, \mathbf{b} \in J_S$ coprime; that arrow exists only for $y \in \mathbf{b}Y_S$, and fullness forces $\mathbf{b}Y_S = Y_S$, i.e. $\mathbf{b} = 1$. So these

bisections are exactly the $B_{\mathbf{a}} = \{(\mathbf{a}, y) : y \in Y_S\}$, with $B_{\mathbf{a}}B_{\mathbf{b}} = B_{\mathbf{ab}}$. This is Lemma 2.1 of Paper XIII run with $\tilde{c}$ in place of the scaling cocycle: there the value of c had to determine the element g , which is exactly norm-separation; the value of $\tilde{c}$ is g , and no hypothesis is needed. Since $\hat{\theta}$ is injective, $\tilde{c}_2 \circ \theta_*$ is constant on a bisection iff $\tilde{c}_1$ is, so θ_* maps the $B_{\mathbf{a}}$ to the $B_{\pi(\mathbf{a})}$ for a monoid isomorphism $\pi : J_{S_1} \rightarrow J_{S_2}$, which carries the free generators of the free abelian monoid J_{S_1} to those of J_{S_2} : a bijection $S_1 \rightarrow S_2$. It preserves norms because $\log N\pi(\mathbf{a}) = c_2(B_{\pi(\mathbf{a})}) = c_1(B_{\mathbf{a}}) = \log N\mathbf{a}$. Evaluating the intertwining of $\tilde{c}$ on $B_{\mathbf{p}}$ gives $\delta_{\mathbf{p}} = \tilde{c}_1(B_{\mathbf{p}}) = \hat{\theta}(\tilde{c}_2(B_{\pi(\mathbf{p})})) = \hat{\theta}(\delta_{\pi(\mathbf{p})})$: $\hat{\theta}$ maps standard generators to standard generators, so θ is induced by π .

Fibres, simplices, symmetries. θ_* maps $B_{\mathbf{a}}$ to $B_{\pi(\mathbf{a})}$, hence the range $\mathbf{a}Y_{S_1}$ to $\pi(\mathbf{a})Y_{S_2}$, hence the valuation fibres to the valuation fibres, and the fibre over 0, which is G_S , homeomorphically onto its counterpart. An isomorphism of C^* -dynamical systems carries KMS_{β} states to KMS_{β} states; by [3, Thm. 3.6] the simplices are $\text{Prob}(G_S/\Xi_{\beta}^{\perp})$, an affine isomorphism of Bauer simplices is a homeomorphism of the extreme boundaries, and the transition locus is the set of β at which the simplex changes. Finally (5) is part (6) of Theorem 2.5 of Paper XIII: its proof uses only the matched bisections and the torsor structure of the fibres, and nowhere uses norm-separation, which entered that paper only through its Theorem 1.4 — replaced here by gauge-equivariance and Lemma 1.1 — and its Lemma 2.1 — replaced by the $\tilde{c}$ argument above; its Remark 2.6 records that these are the only two entry points. $\square$

Corollary 4.3 (Dedekind zeta). *If $S_i = P_{K_i}$ and Φ is gauge- and σ -equivariant, then the norm multisets $\{N\mathbf{p}\}_{\mathbf{p} \in P_{K_1}}$ and $\{N\mathbf{q}\}_{\mathbf{q} \in P_{K_2}}$ agree with multiplicities, hence*

$$\zeta_{K_1}(s) = \prod_{\mathbf{p}} (1 - N\mathbf{p}^{-s})^{-1} = \zeta_{K_2}(s),$$

so K_1 and K_2 are arithmetically equivalent. The narrow class numbers are not among the conclusions: h_S^+ is recovered by no argument of Paper XIII (its Remark 3.2), and nothing here adds one.

Remark 4.4 (Is the hypothesis reasonable?). Gauge-equivariance is a genuine strengthening and should be declared as such. It is however natural: the gauge action is part of the standard data of a semigroup crossed product, it is what makes $C(Y_S)$ the coefficient algebra, and any isomorphism arising from an isomorphism of the underlying dynamical systems $(Y_{S_1}, J_{S_1}) \rightarrow (Y_{S_2}, J_{S_2})$ is automatically gauge-equivariant. What Theorem 4.2 does not do is deduce the conclusion from σ -equivariance alone, and by Corollary 1.4 no argument through $\mathcal{A}^{\sigma}$ ever will.

5. ASSESSMENT

Statement	Status	Reference
$C(Y_S) = \mathcal{A}^{\mathbb{T}^S}$ unconditionally	true	Lem. 1.1
norm-separated $\iff \mathcal{A}^\sigma = C(Y_S)$	equivalent, not sufficient	Thm. 1.3
the hypothesis can be removed	false	Cor. 1.4
P_K norm-separated for $K \neq \mathbb{Q}$	never	Prop. 1.5
$\mathcal{A}^\sigma = \bigoplus_n \mathcal{A}_n^\sigma$	false: not a direct sum	Prop. 2.1
$\mathcal{A}^\sigma$ is L -graded, $\mathcal{A}_{(0)}^\sigma = C(Y_S)$	true	Thm. 2.2
Property A characterizes $C(Y_S)$	circular: uses γ	Rem. 3.1
Property B characterizes $C(Y_S)$	circular: assumes uniqueness	Rem. 3.2
intrinsic recovery from $(\mathcal{A}, \sigma)$	open	Rem. 3.3
gauge-equivariant reconstruction	true, all K, S	Thm. 4.2
θ induced by a bijection $S_1 \rightarrow S_2$	true a posteriori, not a priori	Thm. 4.2(2)
Frobenius classes, $\{\Xi_\beta\}$ matched	if symmetries intertwined	Thm. 4.2(5)
$\zeta_{K_1} = \zeta_{K_2}$ ($S_i = P_{K_i}$)	true, gauge-equivariantly	Cor. 4.3
$h_{K_1}^+ = h_{K_2}^+$	not claimed	Cor. 4.3

The question was well posed and its answer is a change of shape. Norm-separation cannot be removed from Theorem 1.4 of Paper XIII, because Theorem 1.3 shows it is that theorem's conclusion in disguise; what can be done is to replace σ -equivariance by gauge-equivariance, whereupon the reconstruction holds with the exact strength of Paper XIII's Theorem 2.5 for every number field and every prime set, including the classical case $S = P_K$ that Proposition 1.5 shows was previously excluded — with what was conditional there (the Frobenius classes, when the symmetry actions are not known to be intertwined) still conditional, and the narrow class number still not recovered.

Two questions remain, and the first is the sharp form of the original one. Is $C(Y_S)$ intrinsically characterized inside $(\mathcal{A}_{K,S}, \sigma)$ when $L \neq 0$? Remark 3.3 shows the obvious counterexample is blocked by the Nica–Toeplitz relations, so we suspect the answer is yes and that the mechanism is the rigidity of commuting isometries rather than any spectral argument. Second: the L -graded algebra $\mathcal{A}^\sigma$ of Theorem 2.2 is a new object in this programme, being the first non-commutative fixed-point algebra to appear; whether its own structure — for instance whether it is simple, or its ideal lattice — encodes the relation lattice L , and hence the pattern of coincidences among the norms, we have not examined.

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